Lattice: Signed, Replayable Capability Receipts for Model Runs

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***Abstract*: Production systems built on large language models usually keep a free-text log of a run. Lattice is a capability-first TypeScript runtime. When a signer is configured, a run emits a capability receipt: the request, the route, the constraints, and hashes of the outputs. The body is canonicalized with RFC 8785, matched personal data is masked before signing, and the envelope is signed with Ed25519. The signing key is not a receipt field. Routing is a deterministic function, and a violated output invariant is terminal. An offline evaluation against mocked providers measures signing cost, receipt size, tamper rejection, and replay. Replay binding, a schema-version floor, and a chain-length check are verifier opt-ins. New crew runs also close the receipt chain, so a dropped leaf fails verification. Auditability attaches to signed runs.**

***Index Terms*: capability receipts, verifiable replay, LLM orchestration, software supply chain, canonical JSON, digital signatures, reproducibility, multimodal AI**

I. Introduction

Software that calls large language models has moved from prototypes into systems that approve refunds, triage insurance claims, route field operations, and process medical and recruiting documents. These systems combine text with images, audio, documents, structured outputs, and tool calls, and they increasingly spread a single task across several model providers for cost, latency, privacy, or capability reasons. As the stakes rise, two questions follow every run. First, can an operator or an auditor trust the record of what happened. Second, can anyone reproduce that run later, offline, without trusting the party that produced the record. Today the honest answer to both questions is usually no.

The reason is structural. The dominant integration libraries are provider wrappers. They normalize the shape of a chat or generation call across vendors and they provide convenience around tools, memory, and agents. They do not produce a tamper-evident record of a run, they do not commit to a deterministic and inspectable account of why one model was chosen over another, and they do not let a third party replay the run and check the result against a signed artifact. A free text log is not a proof. It can be edited after the fact, it does not bind the inputs to the outputs, and it carries no signature that a verifier can check with a public key.

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This paper argues that the missing primitive is a per-run, signed, replayable *capability receipt*, and that a runtime can make such a receipt a default byproduct of normal operation rather than an after-the-fact instrumentation task. We build this argument concretely through Lattice, a capability-first runtime SDK written in TypeScript. A developer describes the job, supplies any mix of artifacts, declares the desired outputs and policy constraints, and optionally attaches a contract. Lattice then routes the work deterministically, packs context, prepares artifacts for the chosen provider, enforces output invariants, and emits a receipt on every outcome, success or failure.

Lattice provides four guarantees that together make a run auditable and reproducible.

1) **Deterministic and inspectable routing.** The router is a pure function of the capability catalog, the request, and the policy. It uses hard filters, a stable score, and a deterministic tie break, and it reports the full set of candidates, the rejected candidates, the fallback chain, and a typed list of reasons when no route is possible. There is no randomness and no model call inside routing.
2) **Terminal output invariants.** Output level tripwires such as "must cite an artifact" or "no personally identifiable information" are evaluated by a pure kernel. A violation produces a terminal error that the fallback chain is forbidden to retry, which prevents a system from quietly trying again until a banned output slips through.
3) **Signed capability receipts.** Every run emits a receipt that is canonicalized with RFC 8785, redacted before signing, and signed with Ed25519 inside a Dead Simple Signing Envelope. The receipt binds the run record to a key through a signature that any holder of the public key can check.
4) **Offline verifiable replay.** A receipt and its content-addressed artifacts can be materialized into a replay envelope. Materialization verifies the signature before it loads any artifact, and a property test confirms that reconstructing the run and recomputing the output hash yields the value committed in the receipt.

We make the following contributions. We define the capability receipt as a signing and replay primitive for multimodal model runs, and we show how a redact-then-sign ordering and

a canonical encoding turn it into a verifiable artifact rather than a log line. We show how a deterministic and inspectable router, terminal output invariants, and a content-addressed verification tool compose into a runtime where auditability and reproducibility are default properties. We describe a working implementation that ships publicly with supply-chain provenance, modular adoption paths, a seven-provider parity contract, and a test suite that encodes the guarantees as invariants. We also show how external executors can mint compatible receipts without adopting Lattice's provider or agent runtime. The remainder of the paper covers background and related work, the threat model and goals, the design, the implementation, the evaluation, a discussion, the limitations, future work, and a conclusion.

II. Background and Related Work

A. Provider wrappers and agent frameworks

A first family of systems normalizes model access. The Vercel AI SDK [1] offers a broad provider-agnostic TypeScript toolkit with multimodal generation, tools, and agents. LangChain and LangGraph [2] provide orchestration, sessions, and tracing. LiteLLM [3] is strong routing infrastructure that exposes a common interface across many providers with fallback and gateway features. The OpenAI Agents SDK [4] provides agent loops, sessions, and tool-use. The Model Context Protocol [5] standardizes how tools and context attach to a model session. These systems are valuable and widely used. None of them, however, emit a signed and replayable per run receipt. Their tracing is observability for humans, not a cryptographic artifact that a third party can verify against a public key, and their routing is generally either configuration driven or model chosen rather than a deterministic pure function whose full decision is recorded.

B. Software supply-chain provenance

A second family secures the provenance of software artifacts. in-toto [6] provides end-to-end guarantees over a software supply chain by signing the steps that produced an artifact. The Dead Simple Signing Envelope [7] defines a minimal envelope and a pre-authentication encoding so that signatures commit unambiguously to a payload and its type. Sigstore [8] makes signing and transparency broadly accessible. SLSA [9] defines levels of supply chain integrity, and The Update Framework [10] addresses survivable key compromise in update systems. Lattice borrows directly from this lineage. It reuses the DSSE envelope and pre-authentication encoding, it adopts a key set with explicit key states in the spirit of key rotation and revocation, and it treats the published package itself as a supply chain artifact with provenance attestations. The novelty is to apply this provenance discipline not to a build, but to an individual model run at runtime.

C. Canonicalization and signatures

Signing a structured object requires a deterministic byte encoding, otherwise two semantically equal objects can produce different signatures. The JSON Canonicalization Scheme [11] defines such an encoding for JSON, and the I-JSON message format [12] constrains JSON to an interoperable subset, which matters for how numbers and floats are serialized. Lattice canonicalizes receipts with RFC 8785 and serializes monetary cost as an I-JSON string so that receipts never carry raw floats. Signatures use EdDSA [13], the elliptic curve signature construction that is widely deployed for compact and fast signatures.

D. Transparency and reproducibility in machine learning

A third family documents models and data for accountability. Model Cards [14] and Datasheets for Datasets [15] standardize disclosure about a model or dataset, and reproducibility programs in the research community [16] push toward runs that others can repeat. These efforts describe an artifact or a study. A capability receipt is complementary and finer grained. It describes a single production run and makes that run independently checkable and replayable. Merkle trees [17] provide a standard way to commit to a set of items with a single root, which we identify as the natural mechanism for a future signed lineage root over the artifacts that a run consumed and produced.

E. Signed receipts for agent actions

Four systems from 2026 target evidence for autonomous agent runs and are the closest prior work. Sello [18] has the service that receives an agent's call sign a receipt with its own Ed25519 key inside a COSE envelope, encrypt it to the agent's owner, and publish it to a witnessed Merkle log; verification needs the log inclusion proof and a key registry, inputs and outputs appear only as hashes, and the protocol is independent of models and providers. Agent Flight Recorder [19] is middleware that hash-chains agent events into Merkle epochs and anchors epoch roots on a public chain; there is no per-event signature, a verifier needs read access to the chain, and policy verdicts are recorded rather than enforced. NovaFabric [20] captures traffic to several providers into an operator-signed evidence bundle, with Ed25519 at the collector and an ECDSA P-256 seal per tenant inside a DSSE envelope, that is designed for offline verification with stock tools and supports four replay modes; its policy gates act on promotion, export, and replay rather than on live runs, and its author describes the implementation as experimental. Janus [21] is the closest in spirit: a single-node daemon appends each agent step's proposal, gate verdict, and validator response to a hash-chained log whose segments it signs with Ed25519 before the step runs or its effect is released, an auditor re-derives every verdict offline from the log and one public key, and replay re-derived all 287 terminal sagas in the paper's experiment; it does not choose models or providers, model decision records are self-reported rather than signed, and effects on its SDK path are admitted cooperatively rather than held. TrustTrack [22] is a vision paper that proposes verifiable identity, policy commitments, and tamper-resistant behavioral logs for multi-agent systems, without an implementation. Certificate Transparency [23], [24] is the reference design for the log-based

approach that Sello and Agent Flight Recorder build on: it makes issuance auditable after the fact through an append-only log, whereas a Lattice receipt is checkable on its own with a public key and binds the run to a policy decision and a replay path. Table I compares the four implemented systems with Lattice.

F. The gap

Across these families, and to our knowledge, Lattice is the first open SDK that combines per-run signed receipts, deterministic routing, run-time policy enforcement, and offline replay in one runtime. The provider wrappers and agent frameworks route and orchestrate but do not sign; the supply chain systems sign and verify offline but know nothing about model runs; and the agent receipt systems in Table I sign and verify evidence, with Janus also gating effects and replaying decisions, but none of them chooses the model or spans providers, so their evidence is never bound to a routing decision. The design dates from the project's first public commit on April 21, 2026, and the repository history is public.

III. Threat Model and Goals

A. Actors and trust assumptions

We consider four actors. The *issuer* is the Lattice runtime that performs a run and signs a receipt with a private key. The *provider* is an external model service that the runtime calls and that we treat as untrusted and possibly buggy. The *verifier* is any party that later receives a receipt and wishes to check it, possibly without trusting the issuer at the time of verification. The *adversary* may sit between these actors or may hold material that looks legitimate.

B. Adversary capabilities

We assume an adversary who can tamper with a stored receipt, a transcript, or a claimed output after a run completes. We assume an adversary who can attempt to substitute a different model route while presenting a receipt that attests to the intended route. We consider an adversary who has obtained a valid signing key and who then tries to mint a receipt in an older schema in order to bypass integrity fields that newer schemas commit to, a schema downgrade attack. We consider an adversary who replays a valid receipt from a different run, a different relying party, or an earlier time and presents it as proof of the current run. We also consider a careless issuer whose run inputs contain secrets or personally identifiable information that must never be embedded in a signed and shareable artifact.

C. Security goals

The runtime targets five goals. First, integrity of the run record: a receipt that has been altered must fail verification. Second, non-repudiation through signature: a verifier with the public key can confirm which key signed a receipt and whether that key is active, retired, or revoked. Third, redaction before signing: personal data matched by a detector, and the tripwire message that would otherwise quote it, are reduced to a detector name, a length, and a mask before the canonical bytes are produced. The signing key and the provider credential are not receipt fields. A free-text field the caller chooses to store is the caller's responsibility. Fourth, determinism for replay: a verifier can recompute the output hash from the reconstructed run and compare it to the value the receipt commits to. Fifth, downgrade resistance: a receipt in the original schema, or below a caller-supplied `minVersion` floor, is rejected before any cryptographic work. Without that floor, a v1.1 receipt still verifies.

D. Non goals

Several properties are explicitly out of scope. We do not provide confidentiality of provider traffic, which is the responsibility of transport security. We do not reproduce the live behavior of a remote model, since model providers are nondeterministic and outside our trust boundary. Replay reconstructs the recorded run and its output hash, not a fresh call to the provider. A signature alone proves integrity and attribution, not freshness: binding a receipt to a relying party, a nonce, or a validity window requires the optional v1.4 fields and the matching verifier options described under verification below, and a verifier that must accept a receipt only once keeps its own record of accepted content identifiers. Dropping leaf receipts from an open receipt tree leaves a smaller tree that still verifies. The chain verifier then returns an advisory unless the caller passes an expected count. A crew minted by this revision writes a closing receipt that lists member content identifiers, so a dropped leaf fails closed without an out-of-band count. We also do not operate a hosted control plane. Lattice is a runtime SDK, and the receipt is a portable artifact that travels with the run rather than a record locked inside a service.

IV. Design

Figure 1 shows the end-to-end run lifecycle that produces and then consumes a capability receipt. The subsections that follow walk through each stage.

A. The capability-first run model

A run is described by a small intent: a `task` string, a set of `artifacts` that may be text, JSON, images, audio, documents, or tool results, a map of declared `outputs` with their schemas, a `policy` that expresses privacy, latency, provider allow and deny lists, and budget, and an optional `contract`. The developer does not select a model. The runtime selects one from a capability catalog and records the decision. This capability-first shape is what makes a receipt meaningful, because the receipt can record the declared intent and the chosen route side by side.

B. Deterministic and inspectable routing

Routing is implemented as a pure function, `routeDeterministically`, that takes the capability catalog, the request, and the policy and returns a route decision. It performs no input or output, draws no randomness, and never calls a model. The function first applies hard filters that

TABLE I
AGENT RECEIPT SYSTEMS COMPARED WITH LATTICE. CELLS SUMMARIZE EACH PAPER'S OWN DESCRIPTION; "NOT STATED" MEANS THE SOURCE DOES NOT SAY.

System	Signer	Offline verify	Replay	Policy checks	Routing	Multi-provider
Sello [18]	receiving service, Ed25519 in COSE	no; needs log proof and key registry	no; hashes only	no; service denials are recorded	no	not model specific
Agent Flight Recorder [19]	none per event; ECDSA on the chain anchor	no; needs chain read	partial; forensic reconstruction	no; verdicts logged	no	not stated
NovaFabric [20]	operator; collector Ed25519 and tenant ECDSA seal	claimed; not independently tested	yes; four modes	partial; gates on export and replay	no	yes, capture
Janus [21]	log daemon and participants, Ed25519	yes; log and one public key	yes; re-derives decisions	yes; pre-execution and pre-release gates	not stated	not stated
Lattice	runtime signer over a key set, Ed25519 in DSSE	yes; public key only	yes; offline output hash match	yes; invariants, budgets, and policy at run time	yes; deterministic	yes; seven providers

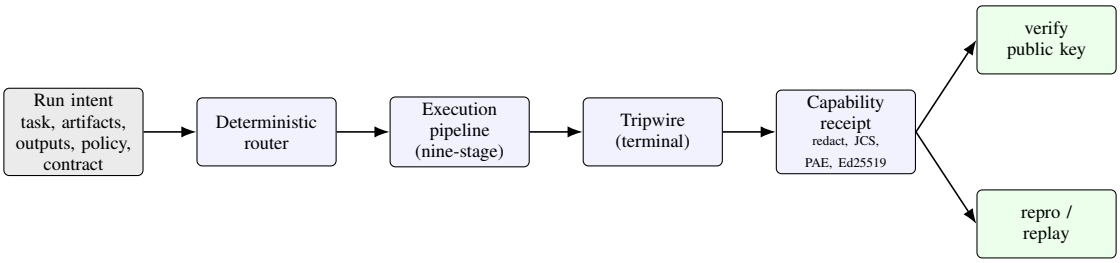


Fig. 1. The Lattice run lifecycle. A run intent is routed by a deterministic and inspectable function, executed through a nine-stage pipeline, checked by terminal output invariants, and committed to a signed capability receipt. The receipt is later consumed by offline verification and by content-addressed reproduction.

reject a candidate when the provider is unavailable, when an explicit provider or model override does not match, when a required input or output modality is unsupported, when structured output is required but unsupported, or when the estimated input exceeds the context window. It then applies policy filters that reject on an allow or deny list, on a privacy class, on a no-logging or no-upload requirement, on a latency class, or on a budget ceiling. Surviving candidates are scored from cost, latency, and context headroom, and ties are broken by a stable order over acceptance status, score, provider name, and model name, with names compared by code unit rather than by locale so the order is the same on every machine. The catalog position is deliberately not part of the score, so the decision does not depend on the order in which providers were registered; a test evaluates every permutation of a near-equal-cost catalog and requires the same selected route and the same fallback chain. The decision exposes the selected route, all candidates, all rejected candidates, a fallback chain of policy-preserving alternatives, and a deduplicated list of typed reasons when no route is possible. The full execution plan is stable JSON with a nine-stage pipeline: analysis, transforms, context packing, provider packaging, tool execution, execution, validation, tripwire, and persistence. Because the decision is a pure function and is recorded in full, the receipt can attest to exactly why a model was chosen, and a reviewer can reproduce the decision by re-running the function on the same inputs.

C. Output invariants with terminal semantics

Lattice lets a developer attach output invariants through a fluent builder, `inv`, with constructors `mustCite`, `fieldFromTable`, `noPII`, and `matches`, where the last validates against a Standard Schema [25]. These invariants are evaluated by a pure kernel, `evaluateTripwires`, that performs no input or output, reads no clock, and uses no randomness, so the same output and the same invariants always yield the same result. The kernel aborts on the first violation. The default detector set for `noPII` contains four detectors: email, United States social security number, credit card validated by the Luhn checksum, and United States phone number. The kernel scans strings and walks nested objects, arrays, and wildcard paths in a deterministic order, so a value buried inside a structured output is not missed. When a no personally identifiable information invariant fires, the runtime error returned to the caller carries the detector name and the matched substring so the violation can be debugged, and the receipt redaction policy replaces that substring with the detector name, the match length, and a shape mask before anything is signed; the other three invariant kinds have their observed value reduced to a type label from a closed set in the same step, and every kind's message is rebuilt from the invariant kind and the declared path so that neither model produced object keys nor schema issue text reach the signed bytes. A violation produces a `TripwireViolationError` that carries the flag `terminal: true`. The predicate `isTerminal` returns true for a tripwire violation and for a no contract match, and the fallback chain

consults this predicate so that a terminal failure is never retried. This matters for safety: a system that retries until a banned output passes is worse than a system that fails closed, and terminal semantics make failing closed the default.

D. Capability contracts and preflight

A contract expresses requirements that must hold before and around a run: a budget with a maximum cost and, for agent runs, iteration and wall time caps, a set of invariants, a quality floor, a set of required modalities, and a required privacy class. A pure preflight evaluator, `evaluateContractAgainstRoute`, runs inside the router and can reject a route with a typed reason such as a budget exceeded, a missing modality, or a privacy mismatch. A quality floor reason is reserved for the evaluation tool described later. Usage is normalized to prompt tokens, completion tokens, and a cost in United States dollars on every outcome, success or failure, and a null cost is distinct from a zero cost so that an unmeasured price is never confused with a free call.

E. The capability receipt

The receipt is the central artifact. Its construction enforces a strict ordering that is the heart of the security argument. Figure 2 shows the pipeline, and Figure 3 shows the resulting envelope.

First, the runtime assembles the raw receipt body from the run record. Second, it redacts the body. Redaction rewrites the tripwire evidence for every invariant kind, replacing a matched personally identifiable value with a detector name, a length, and a shape mask and reducing other observed values to a type label, and records a manifest of redactions, each with a path and a reason, and a signed redaction policy identifier whose default value is `lattice.default.v1`. Each manifest entry is written at the moment of the corresponding rewrite, so the manifest cannot claim a redaction that did not happen. Third, it canonicalizes the redacted body with RFC 8785 [11]. Fourth, it base64 encodes the canonical bytes as the payload. Fifth, it builds the DSSE pre authentication encoding over the payload type and the payload. Sixth, it signs the pre-authentication encoding with Ed25519. The ordering is enforced structurally, not by convention: the canonicalization function is only ever called on the output of the redaction function, so a code path that signs unredacted bytes is unrepresentable. This is what we mean by redact-then-sign as an architectural invariant rather than a guideline.

The signing key is identified by a `kid` and resolved through a key set. A key set maps a `kid` to a key entry whose state is *active*, *retired*, or *revoked*. Cost is serialized as an I-JSON [12] string or null so that receipts never carry raw floating point numbers. The signature is computed over the pre-authentication encoding rather than over the raw JSON, which removes any ambiguity about what the signature commits to.

F. Schema versioning and downgrade defense

Receipts carry an explicit version. Five versions exist: `lattice-receipt/v1`; v1.1, which adds the step-marker

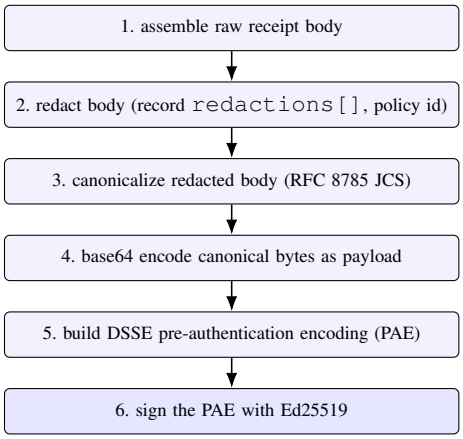


Fig. 2. The receipt construction pipeline. Canonicalization is only ever applied to the output of redaction, so signing unredacted bytes is structurally impossible.

```
{
  "payloadType":
    "application/vnd.lattice.receipt+json",
  "payload": "<base64(JCS(redacted body))>",
  "signatures": [
    { "keyid": "<kid>",
      "sig": "<base64(Ed25519(PAE))>" }
  ]
}
```

Fig. 3. The Dead Simple Signing Envelope produced for a receipt. The pre authentication encoding concatenates the literal `DSSEv1`, the payload type and its length, and the payload and its length, all in a fixed order.

fields; v1.2, which adds an optional and signed model class field recording the training class of the chosen model; v1.3, which adds the parent receipt content identifier and the artifact lineage root; and v1.4, which adds optional audience, nonce, and expiry fields for freshness binding. New receipts are minted at v1.4 and verifiers accept v1.1 and later. Verification rejects a body whose version is the original v1 or is absent, with the error `schema-version-too-low`, and it does so before any cryptographic work. A verifier that depends on a newer schema's commitments raises the floor with a minimum version option and receives the same error for anything older. This is the downgrade defense from the threat model. An adversary who holds a valid signing key still cannot mint a v1-shaped body to evade the integrity fields that later schemas commit to, because the structural version check fails first.

G. Verification

Verification, `verifyReceipt`, returns a result that is either an ok value with the verified body and the resolved key state, or a failure value with a typed error. There are exactly ten error kinds, listed in Table II. Verification checks the envelope shape, the schema version against the floor, the key existence and state, the re-canonicalization match, and the Ed25519 signature, with a defense in depth check that the `kid` inside the body matches the key entry. Only after the signature holds does it apply the optional binding checks: an expected audience, an expected nonce, the receipt's own expiry

TABLE II
THE TEN VERIFYRECEIPT ERROR KINDS, IN THE ORDER THEY ARE CHECKED.

Error kind	Fires when
envelope-malformed	the envelope cannot be decoded or has no signatures
version-mismatch	the body shape or version is unknown
schema-version-too-low	the body is v1, has no version, or is below the caller's floor
key-not-found	the kid is not in the key set
key-revoked	the resolved key is in the revoked state
canonicalization-mismatch	the re-canonicalized body differs from the signed bytes
signature-invalid	the Ed25519 check fails or the body kid mismatches
audience-mismatch	the verifier expects an audience the receipt lacks or differs on
nonce-mismatch	the verifier expects a nonce the receipt lacks or differs on
receipt-expired	the receipt's expiry instant or the verifier's maximum age has passed

instant, and a maximum age over the issue time. Running the binding checks last means an unauthenticated envelope cannot be used to probe which audience or nonce a verifier expects. A companion function, verifyChain, verifies a set of receipts as a tree linked by parent content identifiers: every link must verify, content identifiers must be unique, all receipts must share one run identifier, exactly one receipt may lack a parent, and every receipt must be reachable from that root. Those chain failures are a separate VerifyChainErrorKind union (an invalid link, no root, multiple roots, an orphan link, a run-identifier mismatch, a duplicate content identifier, a count mismatch, or a root mismatch). They are not among the ten VerifyErrorKind values in Table II.

H. Offline verifiable replay

A receipt becomes useful for reproduction through replay. A replay envelope carries optional receipt and contract fields. The function materializeReplayEnvelope reconstructs a replay envelope from a receipt and a set of content-addressed artifacts, and it enforces a critical ordering: it verifies the receipt before it loads any artifact. A test confirms that the artifact loader is called zero times when verification fails, which prevents a tampered receipt from causing artifact loads or side effects. Materialization can fail with one of three kinds: verify-failed, artifact-load-failed, and envelope-malformed. A property test asserts that the round trip from createReceipt through materializeReplayEnvelope through replayOffline preserves the output hash, which is the concrete statement of the determinism goal.

I. The four guarantees and their enforcement

Table III maps each guarantee to the mechanism that provides it and to the property or test that enforces it. The point of the table is that none of the guarantees rest on a convention or a comment. Each is enforced by structure or checked by a test that fails the build when the property is violated.

TABLE III
EACH GUARANTEE, ITS MECHANISM, AND ITS ENFORCEMENT.

Guarantee	Mechanism	Enforcement
Deterministic routing	a pure routing function	recorded decision, no I/O or randomness
Terminal invariants	isTerminal consulted by fallback	fallback never retries a terminal error
Redact-then-sign	canonicalize only on redaction output	ordering is structurally unrepresentable otherwise
Verifiable replay	verify before load	loader call count is zero on verify failure; output hash round trip

V. IMPLEMENTATION

A. Packaging and footprint

Lattice publishes a core runtime package and a command-line package. The core package has a deliberately small runtime dependency surface of four packages: canonicalize for RFC 8785 encoding, @standard-schema/spec for schema contracts, mime for artifact typing, and @noble/ed25519 for one audit signing path. The schema library used in tests is a development dependency only, so a consumer of the core package does not inherit it. Modular public subpaths exist for providers, audit, context, artifacts, routing, tools, storage, eval, agents, and core. These entry-points let a downstream runtime adopt only the provider layer, only signed audit artifacts, or only context and artifact preparation. Package names, the publication workflow, and the Node version used for the measurements are in Appendix XI.

B. Signing internals

Signing uses the Node 24 WebCrypto interface with the literal algorithm name Ed25519. Keys are represented as JSON Web Keys of the octet key pair type and signatures are 64 bytes as specified by EdDSA [13]. To guard against a silent regression in the platform implementation, a development-only parity oracle signs the same message with an independent Ed25519 library and asserts a byte-for-byte match in the test suite. The oracle is never on the production signing path.

C. The seven-provider parity contract

Lattice defines seven logical provider adapters: openai, openai-compatible, anthropic, gemini, xai, openrouter, and lm-studio. An internal invariant named INV-03 states that every improvement must work equally across all seven universal provider targets. A parity smoke test makes this concrete. As summarized in Table IV, the test iterates all seven adapters against provider-shaped fake responses and asserts a uniform adapter shape, string raw outputs, the normalized usage shape, an error on a non-success response, propagation of an abort signal into the underlying fetch, a raw response pass through, and that the set of adapter identifiers has size seven. The parity contract is what lets a receipt mean the same thing regardless of which provider served the run.

TABLE IV
WHAT THE INV-03 PARITY SMOKE TEST ASSERTS ACROSS ALL SEVEN PROVIDERS.

Property	Assertion
adapter shape	each adapter exposes the same surface
raw outputs	content is returned as strings keyed by output name
usage shape	normalized prompt and completion tokens and cost
error path	a non-success response throws with provider identity
cancellation	the request abort signal reaches the fetch
raw response	the original parsed body is available
identity	the set of adapter ids has size seven

The same provider contract now exposes opt-in native provider features for provider only consumers. `ProviderRunRequest.nativeTools`, `nativeToolChoice`, and `nativeStructuredOutput` let an external orchestrator use native tool calling and structured output where adapters support it, without changing the default `ai.run` or agent behavior.

D. The capability registry

Routing consults a capability catalog backed by a registry of 332 model profiles. Of these, 328 are generated from a snapshot of an OpenRouter model feed and 4 are hand maintained static supplements for direct adapters, namely an Anthropic Opus profile, a Gemini Pro profile, an LM Studio local template, and an xAI Grok profile. A weekly continuous integration drift gate guards the generated file so that the catalog does not silently diverge from the upstream feed. At call time the runtime can refine a profile through capability negotiation, fetching a live model listing with a time to live cache and inflight request coalescing, and falling back to the registry on a transient error. Adapter quirk flags capture provider-specific behavior so that the runtime can adapt without breaking the parity contract.

E. External audit and standalone adoption

The receipt layer can also be used without Lattice execution. The `createExternalExecutionAudit` helper accepts externally produced request, response, output, usage, and verdict data, then mints the same signed receipt family, sidecar, replay envelope, and raw request, raw response, input, and output hashes that native Lattice runs produce. Failed verdicts preserve sidecar and receipt evidence but omit replayable outputs, so a failed external execution remains auditable without becoming an offline success fixture.

For runtimes that want preparation without provider execution, `prepareCoreRun` returns artifact references, optional storage references, a context pack, advisory routes, input hashes, warnings, and an execution plan. The tools subpath also exposes MCP resource, prompt, and tool-result artifact helpers, so tool systems can convert their outputs into replayable artifacts without importing the agent runtime.

F. The command-line tool

The `lattice` command-line tool ships in the second package. It uses a lazily loaded subcommand structure. Status codes are listed in Appendix XI. The `verify` subcommand checks the signature and structure of a receipt and prints the key identifier and contract verdict. The `repro` subcommand verifies a receipt, materializes the replay envelope, replays offline, and diffs the recomputed output hash. The `eval` subcommand walks receipt directories or agent snapshots and gates determinism, cost, and quality regressions relative to a baseline, with a disk-backed judge cache. The `receipt diff` subcommand compares signed receipt envelopes and reports structural differences. Together these commands turn a receipt from a passive record into an executable regression and audit gate.

G. Agents and chained receipts

Lattice exposes a runtime agent entry point, `ai.runAgent`, that drives a uniform tool-use loop across all seven providers, and an opt-in multi-agent crew surface, `defineAgent` and `runAgentCrew`. A crew shares a budget with structural caps on total iterations, per-agent iterations, concurrent children, and depth, and it coordinates rate limits through a token bucket with sensible defaults. Crucially, each child receipt chains to a crew root through a parent receipt content identifier, so a multi-agent run produces a verifiable tree of receipts rather than a single opaque trace. Each completion receipt commits to that agent's output hash, and `verifyChain`, exposed in the command-line tool as `receipt verify-chain`, checks every link, the single root, and the parent linkage of the whole tree. An optional expected count covers the truncation case noted in the non-goals, and an optional expected root identifier detects a replaced root. A helper, `evalAgentRun`, gates iterations to goal and cost against a baseline. Three runnable examples exercise these surfaces end to end: a multimodal work inbox that demonstrates a contract, a signed receipt, a tripwire violation, and a no contract match, a single agent loop that emits per-iteration checkpoint receipts, and a crew example that demonstrates chained receipts and an evaluation gate.

VI. EVALUATION

We evaluate Lattice in two ways. The first is measured: the cost that receipts add to a run, the size of a receipt, and whether the verifier rejects every tampered receipt and reproduces every replay, all against a mocked provider so that no model behavior enters the measurement. The second is assurance by construction: the guarantees are encoded as invariants and property tests that fail the build when a guarantee weakens. We do not measure model output quality, which depends on providers outside our trust boundary. A later subsection records a local open-weight model; those runs are not part of the mocked table.

TABLE V
MEASURED RECEIPT COST AND TAMPER RESISTANCE AGAINST A MOCKED PROVIDER. LATENCIES ARE WALL CLOCK; N IS THE NUMBER OF SAMPLES.

Measurement	Result
Create a receipt: redact, canonicalize, sign	median 0.067 ms, p95 0.095 ms (n=1,000)
Verify a receipt	median 0.126 ms, p95 0.177 ms (n=1,000)
Receipt issuance overhead on a run with a 1 MiB image input	1.402 ms, median of 40 signed minus 40 unsigned runs
Envelope size	993 bytes with no inputs, plus 89 bytes per input hash (signed body 593 bytes, plus 66.875)
Tamper sweep	3,744 of 3,744 mutations rejected: 72 receipts, 52 mutations each across 14 classes
Offline replay	350 of 350 runs reproduce the signed output hash

A. Measured cost and tamper resistance

A benchmark script in the repository (`scripts/bench-receipts.mjs`) runs against the built package through its public exports with the in-repo mocked provider and no network, and writes every number below to a committed JSON file (`paper/bench/results.json`) so the table can be regenerated on any machine. Table V reports that run. The Node version and processor are in Appendix XI.

Creating a receipt costs well under a tenth of a millisecond because the body is small and Ed25519 signing is fast; verification is slower than signing because it also re-canonicalizes the body and checks the ten failure branches in order. With a 1 MiB image on the run, issuance adds about half a millisecond, which is the SHA-256 over the input bytes for the input hash and the lineage root. A receipt with no inputs is under a kilobyte on the wire, and each input adds one 64-character hash. The tamper sweep applies 52 mutations to each of 72 signed receipts: single-bit flips at random positions in the payload and in the signature, an unknown key identifier, a valid but wrong key identifier, a swapped body key identifier, a downgrade to the original schema version, a removed version, a dropped output hash, edited audience, nonce, and expiry fields, reordered keys in the payload, a base64url payload, and a payload swapped in from another valid receipt. Every mutation is rejected by a typed verification failure and none makes the verifier throw. The replay check runs 350 runs with distinct outputs through the runtime with a signer, materializes each receipt through the artifact loader, replays it offline, and compares the recomputed output hash with the signed one; all 350 match.

B. Guarantees also encoded as tests

The measurements above are the evidence for cost, tamper resistance, and replay. The repository also contains tests that fail the build if selected guarantees regress. A test count is not itself a measurement, so this subsection only names what those tests check.

We organize those checks around the four guarantees.

Determinism of routing and invariants. The router and the tripwire kernel are pure functions. Tests assert that the same inputs always produce the same route decision and the same tripwire result, that every permutation of a near-equal cost catalog yields the same selected route and fallback chain, and that the no personally identifiable information walk reports the same violation regardless of key insertion order; the absence of input, output, clock reads, and randomness in these functions is verified by construction and by review. Because the decision is recorded in the plan and the receipt, a reviewer can reproduce a route by re-running the function.

Redact-then-sign. The ordering is structurally enforced because the canonicalization function is only ever called on the output of the redaction function. Tests confirm that the redaction manifest and the signed redaction policy identifier appear in the signed body, that for every invariant kind a sentinel value placed in the evidence does not appear anywhere in the canonical signed bytes, that a no personally identifiable information violation is signed as a detector name, a length, and a shape mask while the runtime error still carries the match, and that redaction is idempotent.

Downgrade resistance and verification. Tests cover all ten `VerifyErrorKind` values, including audience, nonce, and expiry mismatches and the caller-raised version floor. Chain failures are not among those ten. They use `VerifyChainErrorKind`. In particular, a body in the original v1 schema, or a body with no version, is rejected with a schema version too low error before any cryptographic work, which demonstrates the downgrade defense. Tests also confirm that a revoked key yields a key revoked error and that a mismatch between the body key identifier and the key entry yields a signature invalid error.

Verifiable replay. A test asserts that the artifact loader is called zero times when receipt verification fails during materialization, which shows that a tampered receipt cannot trigger artifact loads. A property test asserts that the round trip from receipt creation through materialization through offline replay preserves the output hash, which is the determinism goal stated as a checkable property. The `repro` subcommand exercises the same path end to end and returns a distinct exit code for a match and for a drift.

Parity. The INV-03 smoke test exercises all seven providers and asserts the uniform behavior summarized in Table IV, including that the set of adapter identifiers has size seven. Parity is what lets a single receipt format and a single verification path apply regardless of which provider served the run.

A change that weakens one of these checks fails the suite. The checks keep the guarantees from drifting. They do not replace the measured results.

C. Local open-weight runs and other record systems

The artificial evaluation mocks the provider. This subsection does not. Ollama 0.40.7 served `llama3.2:3b` (Llama 3.2, 3.2B, Q4_K_M, digest `a80c4f17acd5`, 2.0 GB) on the same host as the other measurements (Node v24.22.1, 4 vCPU,

15.63 GB RAM, no GPU). Lattice called it through the OpenAI-compatible adapter with no API key. The adapter does not send temperature or a token cap; a fetch wrapper added both, and that fact is part of the record. No vision model was loaded. No hosted provider key was present.

On the one-line prompt "Reply with the single word pong," temperature 0, five paired trials, the median plain HTTP call was 278 ms. A signed Lattice run of the same task was 559 ms and an unsigned Lattice run was 745 ms (Figure 4). The first plain call, 4845 ms, is a cold start and pulls the mean up; the median does not. The signed median is lower than the unsigned median. That gap is generation noise, not a claim that signing is faster. What is stable is the time spent outside the provider HTTP call: 0.439% of end-to-end latency at the signed median (minimum 0.335%, maximum 2.469%). The Lattice prompt body was 382 characters against a one-line plain prompt, so most of the extra wall time is a larger prefill, not the signature. The short signed receipt was 996 bytes.

A refund-policy question with a `mustCite` invariant passed on all three temperature-0 runs and all three temperature-0.7 runs. Each temperature produced one identical answer across its three trials (n=3). The two temperatures did not produce the same sentence. All 6 signed receipts verified, replayed offline, matched the output hash, and a one-character edit was detected (6/6). A synthetic support ticket containing an email, a US phone number, and a dashed SSN was flagged by those three detectors. The model's answer contained none of them, the `noPII` invariant passed, and the signed receipt did not contain the planted strings. A two-step agent called `lookup_policy` once and answered; the returned agent object carried no receipt. A two-agent crew completed in 3 iterations with 4 receipts.

On the public sets the same detectors are weaker than on the synthetic harness, and the gap is the result (Figure 5). ai4privacy/pii-masking-300k English validation (revision `c8c77895a005`, 7,946 documents; academic, non-commercial, no redistribution) gives email precision 0.955 and recall 0.979 (1533 documents with a gold email). US SSN precision is 0.325 and recall 0.436. US phone precision is 0.316 and recall 0.333. Of the SSN false positives, 1,074 overlap a passport span: a 9-digit passport is the same shape as an SSN, and the detector cannot tell them apart. Of the phone false positives, 629 overlap a social-security span. There is no gold credit-card span in this file; the card detector fired on 7 documents anyway. Among 1,070 US-shaped SSN spans, the single match the API returns overlaps 663. It misses 370 because that one match fell on a different number earlier in the document, and 37 because it matched nothing. Gretel English finance test (Apache-2.0, revision `7b844d167385`, 2,891 rows) is the same pattern at smaller n: email recall 0.934, SSN precision 0.161. ASQA dev (Apache-2.0, revision `084060f16b46`, 948 answers) is a loss for `mustCite`. A gold Wikipedia title occurs as a substring in 525 answers, but the invariant accepts at least one gold title in 7. Only 2 answers use a square-bracket citation. The check is syntactic. It does not score whether a citation supports the claim.

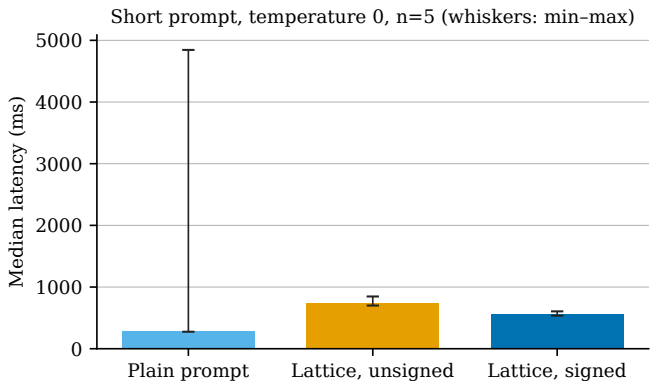


Fig. 4. Median latency of a one-line completion. Whiskers run from the minimum to the maximum of five trials. The plain arm sends only that line. The Lattice arms send the adapter's packed prompt.

Table VI wraps the same recorded request and response bytes. Lattice's receipt and Sello's encrypted receipt (mock transparency log, not the public log) omit the planted strings. An AFR log with disclosure encryption also omits them; the default AFR log contains them. Janus's local evidence appender stores the payload in the clear and the byte edit fails `verify.SegmentDir`. A sealed NovaFabric capsule (self-asserted ECDSA key, no RFC 3161 timestamp) fails `nova verify` after a newline is appended to stdout, and the capsule contains the planted text. `nova validate` on that edited capsule still exits 0. cosign `sign-blob` with a local key, not Fulcio or Rekor, detects a byte edit and signs the plaintext log. in-toto's DSSE link detects a byte edit and does not embed the plaintext. A hash chain detects an edit that leaves the stored hashes untouched and accepts a rebuilt chain, because nothing outside the file anchors the head. NovaFabric's HMAC record has the same shape with a shared secret: an edit fails, and a re-sign with that secret passes. OpenTelemetry GenAI spans and a plain JSON log have no integrity check; both store the transcript. TrustTrack has no implementation we could run. Langfuse was not started: this host has no Docker daemon. Figure 6 is the byte column.

VII. Discussion

Signed capability receipts change what is possible after a run. An operator can hand a receipt to an auditor who verifies it with a public key and replays it offline, without access to the original system and without trusting the operator. A dispute about what a system produced becomes a check against a signed artifact rather than an argument about log files. A continuous integration pipeline can use the `eval` subcommand to gate regressions in cost, in determinism, and in quality across a corpus of receipts, which makes a model or prompt change a reviewable event with measurable consequences. Because the package itself is published with provenance, the chain of trust extends from the source commit through the published tarball to the receipts that the runtime emits.

TABLE VI
SAME RECORDED OLLAMA RUNS, WRAPPED BY EACH SYSTEM THAT RAN. WRAP TIME EXCLUDES MODEL GENERATION. TRUSTTRACK AND LANGFUSE WERE NOT EXECUTED.

System	Status	Wrap ms	Bytes	Offline verify	Byte edit detected	Key assumption	PII in artifact	Replay of output
Lattice receipt	measured	5.923	3210	yes	yes	other key rejected	no	yes
JSON log	baseline	0.033	1045	no	no	—	yes	no
Hash chain	baseline	0.082	1531	yes	yes	rebuild accepted	yes	no
OTel GenAI spans	measured	0.228	1366	no	no	—	yes	no
Agent Flight Recorder	measured	0.797	2328	yes	yes	mock anchor only	yes	no
AFR disclosure	measured	1.907	3357	yes	yes	mock anchor only	no	no
NovaFabric HMAC	measured	0.077	1432	yes	yes	shared secret re-signs	yes	no
NovaFabric capsule	measured	1373.053	21288	yes	yes	self-asserted key	yes	no
cosign sign-blob	measured	4916.329	1319	yes	yes	local key, no Rekor	yes	no
in-toto link	measured	34.149	888	yes	yes	functionary key	no	no
Sello receipt	measured	9.336	1359	yes	yes	mock log	no	no
Janus evidence log	measured	5	1723	yes	yes	writer key	yes	no
TrustTrack	paper only	—	—	—	—	—	—	—
Langfuse	not run	—	—	—	—	—	—	—

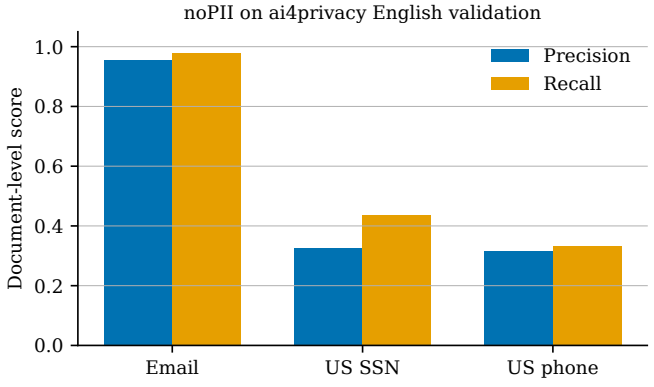


Fig. 5. Document-level precision and recall of the default `noPII` detectors on ai4privacy English validation. US SSN and US phone lose to identifier types that share their shape.

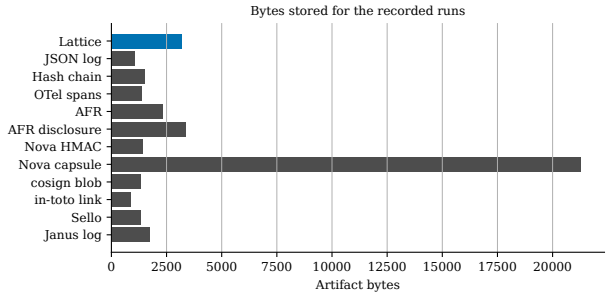


Fig. 6. Bytes retained for the recorded runs. The NovaFabric bar is one sealed capsule, not three hash receipts. AFR disclosure is the encrypted archive.

The design has also been tested from outside. An independent evaluation harness run against release 1.5.1 [26] found three gaps: a violated no-personally-identifiable-information invariant copied the matched substring into the signed evidence while the redaction manifest claimed it had been redacted; receipts carried no freshness or audience binding, so a valid receipt could be replayed to a different relying party at any later time; and crew receipts formed a tree that no verifier checked as a whole. Evidence redaction shipped in the following release [27]: matched personal data is rewritten to a detector name, a length, and a mask before signing. Replay binding (audience, nonce, and expiry) and the caller-supplied schema-version floor remain verifier opt-ins, so an older receipt still verifies. `lattice verify` prints an advisory when those checks were not requested. `expiresAt`, when the issuer set it, is enforced. `verifyChain` checks every link, the single root, and parent reachability. Its failures are a `VerifyChainErrorKind`, not one of the ten `VerifyErrorKind` values. Tail truncation of an open chain is a hard failure only when the caller passes an expected count. Otherwise the result carries an advisory. Crew runs in this revision mint a closing receipt that lists member content identifiers, so a dropped leaf fails verification without that count. The commit those measurements were taken on is recorded in Appendix XI and in `eval/results/env.json`.

Three design tensions are worth naming. The first is determinism against model nondeterminism. Models are not deterministic, so Lattice does not promise to reproduce a live call. It promises to reproduce the recorded run and to bind the inputs to the output hash, and it offers a layered determinism check in the evaluation tool for free form text. The second is redaction against reproducibility. A receipt that redacts a secret cannot reproduce a run that depended on that secret value, so redaction is scoped to material that must never be shared, and the redaction manifest makes the omission explicit and signed. The third is modularity against uniformity. Provider-only consumers can opt into native tools and structured outputs, while the default runtime and agent paths preserve their existing portable behavior. We accept this explicit split because parity is what makes a receipt portable.

VIII. LIMITATIONS

Several limitations follow directly from the design. Replay reconstructs the run record and the output hash, not a fresh model call, so a receipt is evidence about a past run rather than a way to obtain new model behavior. Semantic determinism for free-form text relies on a layered judge in the evaluation

tool, which is more expensive and less crisp than an exact hash match. The capability registry is a point-in-time snapshot refreshed by continuous integration, so a brand new model may be absent until the next refresh, although capability negotiation can refine a profile at call time. When a configured signer fails, the runtime does not hide the gap: by default the result carries a receipt error field and an event, and an operator can instead configure the run to fail. The default personally identifiable information detectors are regular expressions over four United States and email formats documented in the source; names, addresses, and non-United States formats are not detected, and the must-cite invariant checks that a citation is present, not that it supports the claim. No replay cache ships with the verifier, so a relying party that must accept a receipt only once keeps its own record of accepted content identifiers. An open chain warns on tail truncation unless an expected count is passed. A crew closing receipt makes a dropped leaf a count mismatch without an out-of-band count. Finally, external audit helpers can sign and replay evidence produced elsewhere, but they cannot prove that the external executor used the claimed provider unless that executor supplies trustworthy raw envelopes.

IX. Future Work

Several extensions build naturally on the receipt primitive. External audit can become stronger when executors attach provider-side attestations or gateway signed envelopes. The run event vocabulary can be exported more broadly through OpenTelemetry [28] so receipts and traces share a vocabulary across organizations. Key management can be hardened with key management service backed signers and rotation policies. Realtime audio and video sessions need a distinct session surface and a session summary receipt, since a bidirectional stream differs structurally from a request and response. Finally, the modular subpaths can be validated against more downstream runtimes so adoption does not require replacing an existing orchestration stack.

X. Conclusion

Production systems built on large language models need runs that an auditor can verify and an operator can reproduce. We presented Lattice, a capability-first runtime in which a signed multimodal model run emits a capability receipt that is canonicalized, redacted before signing, and signed inside a Dead Simple Signing Envelope, backed by a deterministic and inspectable router and by output invariants with terminal semantics. A verifier can check a receipt with a public key and replay the run offline, confirming that the recomputed output hash matches the signed value. The implementation ships publicly with supply-chain provenance, a seven-provider parity contract, a registry of 332 model profiles, modular adoption paths, a measured sub-millisecond signing and verification cost, and a test suite that encodes the guarantees as invariants. By making a signed and replayable receipt a byproduct of every signed run, Lattice turns auditability and reproducibility from after-the-fact instrumentation into properties of the runtime itself.

XI. Reproducibility notes

The benchmark in `paper/bench/results.json` was produced by `scripts/bench-receipts.mjs` on Node v24.22.1 (linux 6.12.94+ x64), processor Intel(R) Xeon(R) Processor, Lattice 1.6.0, commit `4549334`. Provider: mocked (createFakeProvider), no network.

The packages `@full-self-browsing/lattice` and `@full-self-browsing/lattice-cli` are published on the public npm registry with an OpenID Connect trusted publisher and with supply-chain provenance attestations in the spirit of SLSA [9]. A tagged GitHub release accompanies each publish. The runtime targets Node 24 or newer. Modular subpaths for audit, context, artifacts, routing, tools, eval, and core are also exercised as Node-20-style consumers in the test suite.

The command-line tool returns exit status 0 for success, 1 for a verification or test failure, and 2 for a load or configuration failure. A successful `verify` may print `WARN` lines on standard error when audience, nonce, expiry, or a version floor were not checked. The process still exits 0.

The offline evaluation harness lives in `eval/`. `eval/run-all.sh` rebuilds the packages and writes `eval/results/`, including hardware and the git commit. Provider calls in that harness are in-process fakes. No API key is required.

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